

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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Course Outcome

CO1

1. TITLE

Develop Dynamic Webpage Using JavaScript

2. AIM

To develop a dynamic Student Result System webpage using HTML, CSS and JavaScript that accepts student details, validates marks, calculates total and average, determines pass/fail status and assigns a grade.

3. OBJECTIVE

The objectives of this experiment are:

- To create a dynamic webpage using JavaScript.
- To accept student name and subject marks through input fields.
- To validate empty inputs and mark ranges.
- To calculate total marks and average.
- To determine PASS or FAIL based on subject marks.
- To assign grades based on the average mark.
- To use JavaScript functions and DOM manipulation.
- To implement a Dark Mode / Light Mode toggle.

4. THEORY

JavaScript is a scripting language used to add dynamic behavior and interactivity to web pages. It can access and modify HTML elements through the Document Object Model (DOM).

In this application, JavaScript is used to read values from input fields, validate the entered data, perform calculations, determine the result and dynamically display the result on the webpage.

- `getElementById()` is used to access HTML elements.
- `Number()` converts input values into numbers.
- `classList.toggle()` is used to switch between normal and dark mode.
- `if...else` statements are used for validation, pass/fail checking and grade calculation.
- style properties are used to change the result box dynamically.
- `innerHTML` is used to display the calculated result.

5. ALGORITHM / PROCEDURE

1. Create an HTML document and add the required input fields and buttons.
2. Create input fields for student name and three subject marks.
3. Create a result box to display the calculated result.
4. Define the `toggleMode()` function for Dark Mode and Light Mode.

5. Define the calculateResult() function.
6. Read the student name and subject marks from the input fields.
7. Validate that the student name is not empty.
8. Validate that all subject marks have been entered.
9. Check that every mark is between 0 and 100.
10. Calculate the total and average marks.
11. Check whether all three marks are at least 35 to determine PASS or FAIL.
12. Assign the grade according to the average.
13. Apply dynamic styling to the result box.
14. Display the student name, total, average, status and grade.
15. Execute the webpage in a browser and verify the output using different test cases.

6. PROGRAM

Complete HTML, CSS and JavaScript Source Code:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Result System</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 50px;
    }
    input {
      padding: 8px;
      margin: 5px;
    }
    button {
      padding: 10px 20px;
      cursor: pointer;
    }
    #result {
      margin: 30px auto;
      padding: 20px;
      width: 300px;
      border: 2px solid black;
    }
  </style>
</head>

<body>
  <div>
    <input type="text" value="Student Name" />
    <input type="text" value="Subject 1" />
    <input type="text" value="Subject 2" />
    <input type="text" value="Subject 3" />
    <button value="Calculate Result" />
  </div>
  <div id="result">
    <div>
      <div></div>
      <div></div>
      <div></div>
      <div></div>
      <div></div>
    </div>
  </div>
</body>
</html>
```

```

        border-radius: 10px;
        display: none;
    }

    .dark-mode {
        background-color: #222;
        color: white;
    }

</style>
</head>

<body>

    <button onclick="toggleMode()" id="modeButton">🌙 Dark Mode</button>
    <h1>Student Result System</h1>

    <label>Student Name:</label>
    <input type="text" id="name">
    <br><br>

    <label>Subject 1:</label>
    <input type="number" id="mark1">
    <br><br>

    <label>Subject 2:</label>
    <input type="number" id="mark2">
    <br><br>

    <label>Subject 3:</label>
    <input type="number" id="mark3">
    <br><br>

    <button onclick="calculateResult()">Calculate Result</button>
    <div id="result"></div>

    <script>
        function toggleMode() {
            document.body.classList.toggle("dark-mode");
            let button = document.getElementById("modeButton");

            if (document.body.classList.contains("dark-mode")) {
                button.innerHTML = "☀️ Light Mode";
            } else {
                button.innerHTML = "🌙 Dark Mode";
            }
        }

        function calculateResult() {

```

```

let name = document.getElementById("name").value;

let mark1Input = document.getElementById("mark1").value;
let mark2Input = document.getElementById("mark2").value;
let mark3Input = document.getElementById("mark3").value;

let resultBox = document.getElementById("result");

// Validate name
if (name === "") {
    resultBox.style.display = "block";
    resultBox.style.color = "red";
    resultBox.innerHTML = "Please enter student name!";
    return;
}

// Validate empty marks
if (mark1Input === "" || mark2Input === "" || mark3Input ===
"" ) {
    resultBox.style.display = "block";
    resultBox.style.color = "red";
    resultBox.innerHTML = "Please enter all subject marks!";
    return;
}

let mark1 = Number(mark1Input);
let mark2 = Number(mark2Input);
let mark3 = Number(mark3Input);

// Validate mark range
if (
    mark1 < 0 || mark1 > 100 ||
    mark2 < 0 || mark2 > 100 ||
    mark3 < 0 || mark3 > 100
) {
    resultBox.style.display = "block";
    resultBox.style.color = "red";
    resultBox.innerHTML = "Marks must be between 0 and 100!";
    return;
}

// Calculate total and average
let total = mark1 + mark2 + mark3;
let average = total / 3;
let status;
let grade;
// Check pass/fail
if (mark1 >= 35 && mark2 >= 35 && mark3 >= 35) {

```

```

        status = "PASS";

        if (average >= 90) {
            grade = "A+";
        }
        else if (average >= 80) {
            grade = "A";
        }
        else if (average >= 70) {
            grade = "B";
        }
        else if (average >= 60) {
            grade = "C";
        }
        else if (average >= 50) {
            grade = "D";
        }
        else {
            grade = "F";
        }

        // Dynamic styling for PASS
        resultBox.style.color = "green";
        resultBox.style.borderColor = "green";

    } else {

        status = "FAIL";
        grade = "F";

        // Dynamic styling for FAIL
        resultBox.style.color = "red";
        resultBox.style.borderColor = "red";
    }

    // Show result box
    resultBox.style.display = "block";

    // Display result
    resultBox.innerHTML =
        "<h2>Student Result</h2>" +
        "<p><b>Name:</b> " + name + "</p>" +
        "<p><b>Total:</b> " + total + "</p>" +
        "<p><b>Average:</b> " + average.toFixed(2) + "</p>" +
        "<p><b>Status:</b> " + status + "</p>" +
        "<h2>Grade: " + grade + "</h2>";
    }
</script>
</body>

```

</html>

7. COMPLEX BLOCKS / FUNCTIONS IMPLEMENTATION

7.1 Dark Mode / Light Mode Function

The toggleMode() function dynamically adds or removes the dark-mode class from the body. It then checks the current mode and changes the button text accordingly.

```
function toggleMode() {
    document.body.classList.toggle("dark-mode");
    let button = document.getElementById("modeButton");

    if (document.body.classList.contains("dark-mode")) {
        button.innerHTML = "☀ Light Mode";
    } else {
        button.innerHTML = "🌙 Dark Mode";
    }
}
```

7.2 Input Validation Block

The calculateResult() function first validates the student name and checks whether all subject marks are entered. If validation fails, a message is displayed and return stops further execution.

```
if (name === "") {
    resultBox.style.display = "block";
    resultBox.style.color = "red";
    resultBox.innerHTML = "Please enter student name!";
    return;
}

if (mark1Input === "" || mark2Input === "" || mark3Input === "") {
    resultBox.style.display = "block";
    resultBox.style.color = "red";
    resultBox.innerHTML = "Please enter all subject marks!";
    return;
}
```

7.3 Mark Range Validation

The program converts the input strings into numbers and checks all three marks together. The program stops if any mark is outside the valid range of 0 to 100.

```
let mark1 = Number(mark1Input);
let mark2 = Number(mark2Input);
let mark3 = Number(mark3Input);

if (
    mark1 < 0 || mark1 > 100 ||
    mark2 < 0 || mark2 > 100 ||
    mark3 < 0 || mark3 > 100
) {
    resultBox.style.display = "block";
    resultBox.style.color = "red";
    resultBox.innerHTML = "Marks must be between 0 and 100!";
}
```

```
        return;
    }
}
```

7.4 PASS/FAIL and Grade Calculation

This is the main decision-making block. The student must score at least 35 in every subject to pass. If the student passes, nested if...else conditions assign a grade based on the average.

```
if (mark1 >= 35 && mark2 >= 35 && mark3 >= 35) {
    status = "PASS";

    if (average >= 90) {
        grade = "A+";
    }
    else if (average >= 80) {
        grade = "A";
    }
    else if (average >= 70) {
        grade = "B";
    }
    else if (average >= 60) {
        grade = "C";
    }
    else if (average >= 50) {
        grade = "D";
    }
    else {
        grade = "F";
    }
} else {
    status = "FAIL";
    grade = "F";
}
```

7.5 Dynamic Result Generation

The program makes the result box visible and builds its contents dynamically using innerHTML. The average is formatted to two decimal places using toFixed(2).

```
resultBox.style.display = "block";

resultBox.innerHTML =
    "<h2>Student Result</h2>" +
    "<p><b>Name:</b> " + name + "</p>" +
    "<p><b>Total:</b> " + total + "</p>" +
    "<p><b>Average:</b> " + average.toFixed(2) + "</p>" +
    "<p><b>Status:</b> " + status + "</p>" +
    "<h2>Grade: " + grade + "</h2>";
```

8. HOW TO RUN THE PROGRAM

- Open the HTML file in Visual Studio Code.
- Save the file with a .html extension, for example: student_result.html.
- Open the HTML file using a web browser. You can use Live Server in VS Code if it is installed.
- Enter the student name and marks for all three subjects.

- Click the Calculate Result button to view the total, average, status and grade.
- Use the Dark Mode button to switch between Dark Mode and Light Mode.

9. CODE EXPLANATION

Code / Concept	Purpose
toggleMode()	Switches the webpage between Dark Mode and Light Mode.
calculateResult()	Validates inputs, calculates the result and displays it.
getElementById()	Accesses HTML elements using their IDs.
classList.toggle()	Adds or removes the dark-mode CSS class.
Number()	Converts entered mark values into numbers.
if...else	Performs validation, pass/fail checking and grade selection.
total and average	Calculates the sum and average of the three subject marks.
toFixed(2)	Displays the average with two decimal places.
style.color / borderColor	Changes result box styling according to PASS or FAIL.
innerHTML	Dynamically inserts the student result into the result box.

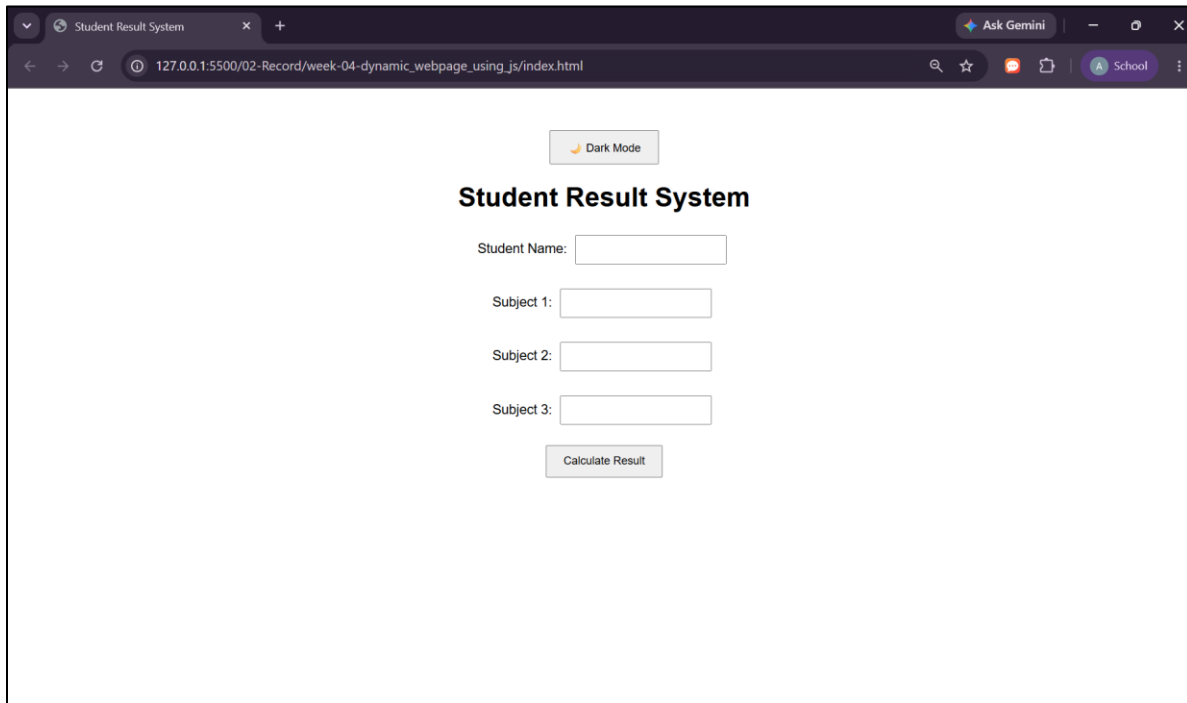
10. TEST CASES AND EXECUTION DATA

The following test cases can be executed to verify the main functionality of the webpage.

Test Case	Input	Expected Result	Actual Result / Status
TC-01	Venu; 85, 90, 80	PASS; Grade A; Total 255; Average 85.00	Test Case Passed
TC-02	Ravi; 95, 92, 96	PASS; Grade A+; Total 283; Average 94.33	Test Case Passed
TC-03	Arjun; 30, 75, 80	FAIL; Grade F; Total 185; Average 61.67	Test Case Passed
TC-04	Name empty; 80, 80, 80	Please enter student name!	Test Case Passed
TC-05	Venu; one mark empty	Please enter all subject marks!	Test Case Passed
TC-06	Venu; 105, 80, 90	Marks must be between 0 and 100!	Test Case Passed

11. OUTPUT

Output 1: Student Result System – Input Page



A screenshot of a web browser displaying the 'Student Result System' input page. The browser's address bar shows the URL '127.0.0.1:5500/02-Record/week-04-dynamic_webpage_using_js/index.html'. The page has a dark theme toggle button labeled 'Dark Mode' at the top. Below it, the title 'Student Result System' is centered. The form contains four input fields: 'Student Name:', 'Subject 1:', 'Subject 2:', and 'Subject 3:'. A 'Calculate Result' button is positioned below the subject fields.

Dark Mode

Student Result System

Student Name:

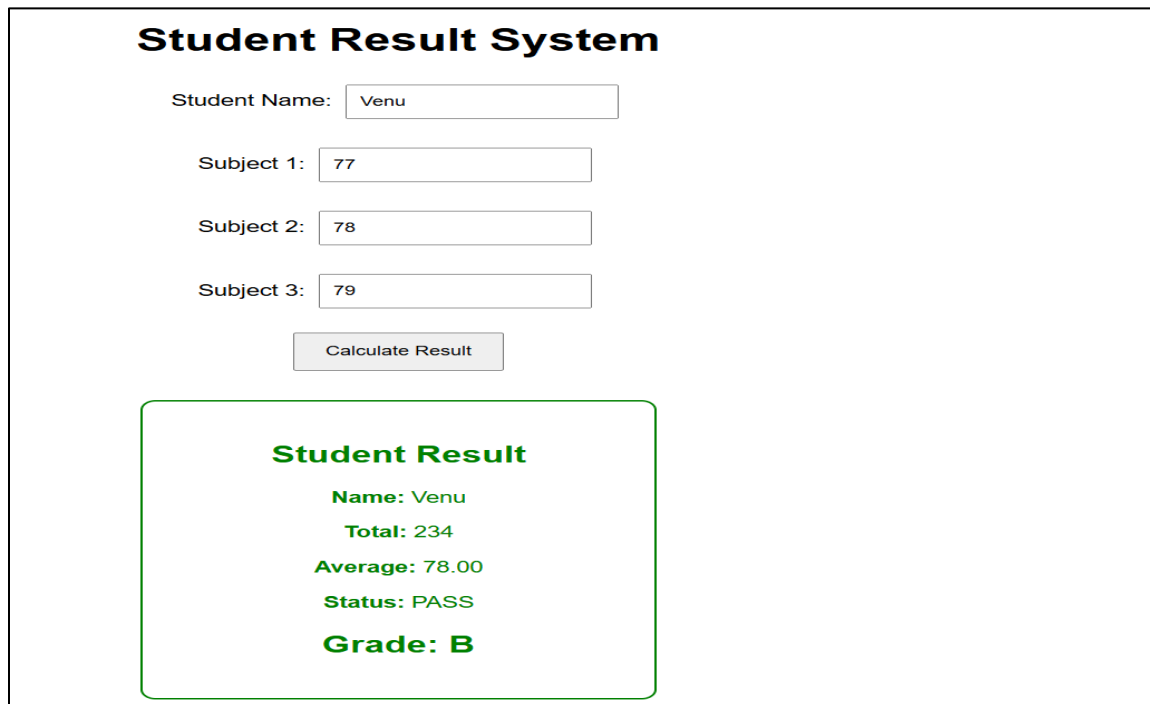
Subject 1:

Subject 2:

Subject 3:

Calculate Result

Output 2: PASS Result



A screenshot of the 'Student Result System' showing the results for a student named Venu. The input fields are filled with 'Venu', '77', '78', and '79'. The 'Calculate Result' button is visible. Below the input fields, a green-bordered box displays the results: 'Student Result', 'Name: Venu', 'Total: 234', 'Average: 78.00', 'Status: PASS', and 'Grade: B'.

Student Result System

Student Name:

Subject 1:

Subject 2:

Subject 3:

Calculate Result

Student Result

Name: Venu
Total: 234
Average: 78.00
Status: PASS
Grade: B

Output 3: FAIL Result

Student Result System

Student Name:

Subject 1:

Subject 2:

Subject 3:

Student Result
Name: Venu
Total: 120
Average: 40.00
Status: FAIL
Grade: F

Output 4: Validation Message

Student Result System

Student Name:

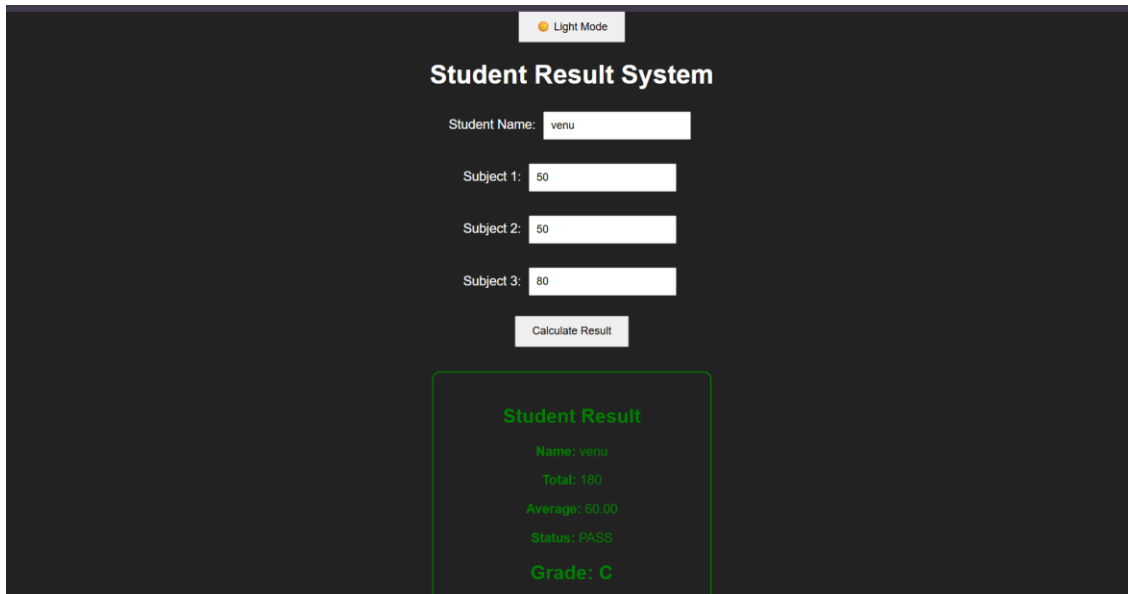
Subject 1:

Subject 2:

Subject 3:

Please enter all subject marks!

Output 5: Dark Mode



Light Mode

Student Result System

Student Name:

Subject 1:

Subject 2:

Subject 3:

Student Result

Name: venu

Total: 180

Average: 60.00

Status: PASS

Grade: C

12. RESULT

Thus, a dynamic Student Result System webpage was developed using HTML, CSS and JavaScript. The application accepts student details, validates the input, calculates total and average marks, determines PASS or FAIL, assigns a grade and displays the result dynamically. Dark Mode and Light Mode functionality was also implemented.

13. CONCLUSION

The experiment demonstrates the use of JavaScript functions, DOM manipulation, conditional statements, input validation, calculations and dynamic webpage styling.